MATH 110 — Week 2

Homework 2 — Solutions

HW2 P1.

Idea. Independence of the two lists says nothing about cancellation between them.

1. In $V = \mathbb{R}$, choose the one-element lists $v_1 = 1$ and $w_1 = -1$. Each is independent since a nonzero scalar in a field has an inverse.
2. Their termwise sum is the list (0), which is dependent because $1 \cdot 0 = 0$ has nonzero coefficient. The assertion is false.

HW2 P2(a).

Idea. Each vector-space identity is a field identity at every coordinate.

1. Addition and scalar multiplication produce sequences in F^∞, so the operations are closed.
2. Addition is commutative and associative coordinatewise. The zero sequence is an additive identity and the negative sequence is the additive inverse.
3. For scalars a, b and sequences x, y, each coordinate satisfies $(ab)x_j = a(bx_j)$, $a(x_j + y_j) = ax_j + ay_j$, $(a + b)x_j = ax_j + bx_j$, and $1x_j = x_j$. Thus all vector-space axioms hold.

HW2 P2(b,c).

Idea. Arbitrarily many independent coordinate sequences rule out finite dimension, but finite combinations do not produce every sequence.

1. For each N, the list $e_1, \ldots, e_N$ is independent: its zero combination has jth coordinate equal to its jth coefficient.
2. If a list of length M spanned F^∞, the independent list of length $M + 1$ would contradict Notes Proposition 10. Hence the space is infinite-dimensional.
3. Any finite combination of coordinate sequences is zero past the largest index used. The sequence $(1, 1, \ldots)$ is not, so the set of all e_j does not span.

HW2 P3.

Idea. A basis extracted from vectors in the union remains in that union.

1. By Proposition 14, the subspaces U, W have finite bases, since V is finite-dimensional.
2. Join these bases. The joined list spans $U + W = V$: expand the two summands separately.
3. Proposition 9 deletes vectors to leave a basis. Every retained vector still lies in U or W, proving the assertion.

HW2 P4(a).

Idea. The only dimension strictly between 0 and 2 is 1.

1. The subspace is nonzero, so a nonzero v gives an independent one-element list, and its dimension is at least 1.
2. Its dimension is at most 2 by Proposition 14. Equality would make it all of F^2, contradicting properness; hence its dimension is 1.
3. Any nonzero v is then a basis by Proposition 13, so the subspace is $\operatorname{span}(v)$.

HW2 P4(b).

Idea. A dependence between two vectors can be solved for one in terms of the other.

1. If v, w are a basis and $v = aw$, then $v - aw = 0$ is a nontrivial relation, contradiction. The same applies if $w = av$.
2. Conversely assume neither is a scalar multiple of the other. Neither can be zero, since zero is a scalar multiple of every vector.
3. If $av + bw = 0$ nontrivially, $a \neq 0$ gives $v = -(b/a)w$; if $a = 0$ then $b \neq 0$ gives $w = 0v$. Both contradict the assumption.
4. The list is independent of length $2 = \dim F^2$, so Proposition 13 makes it a basis.

HW2 P5.

Idea. The dimensions already add to the dimension of the sum.

1. Proposition 16 gives $\dim(U \cap W) = 3 + 5 - 8 = 0$.
2. A zero-dimensional space is $\{0\}$, because its empty basis spans only zero.
3. The intersection is zero and the sum is $\mathbb{R}^8$, so the definition of direct sum gives the conclusion.

HW2 P6(a).

Idea. Evaluation and differentiation preserve the specified zero equalities.

1. The zero polynomial satisfies both conditions.
2. If $p, q \in U$ and $a \in \mathbb{R}$, then $(ap + q)(1) = ap(1) + q(1) = 0$, and $(ap + q)'(1) = ap'(1) + q'(1) = 0$.
3. Thus $ap + q \in U$, proving the subspace property by Week 1's test.

HW2 P6(b).

Idea. Use the polynomial basis centered at the point where values are prescribed.

1. Example 18 gives the unique expression $p(x) = c_0 + c_1(x-1) + \cdots + c_4(x-1)^4$.
2. Evaluation at 1 gives $p(1) = c_0$; differentiate and evaluate to get $p'(1) = c_1$.
3. Thus $p \in U$ exactly when $c_0 = c_1 = 0$. The list $(x-1)^2, (x-1)^3, (x-1)^4$ spans U and is independent, as a sublist of that basis.
4. It is a basis with three members, so $\dim U = 3$. This uses coefficients, not a dimension theorem about linear maps.

Exam practice, Question 1(a).

Idea. Show the four generators are independent, then use the dimension-four basis test.

1. A relation $au_1 + bu_2 + cw_1 + dw_2 = 0$ gives $a + d = 0$, $a + b = 0$, $b + c = 0$, $c - d = 0$.
2. Thus $a = -d$, $b = d$, $c = -d$, and $-d = d$. Over $\mathbb{R}$, this implies $d = 0$, hence all coefficients vanish.
3. The four generators are a basis of $\mathbb{R}^4$, proving the sum is all of it. If a vector belongs to $U \cap W$, equating its U and W combinations gives such a relation, so it is zero. The sum is direct.

Exam practice, Question 1(b).

Idea. Solve the same four coordinate equations with the specified target.

1. The equations are $a + d = 2$, $a + b = 1$, $b + c = 3$, $c - d = 0$.
2. The last equation gives $c = d$; the first two give $a = 2 - d$, $b = d - 1$. Thus $2d - 1 = 3$, so $(a, b, c, d) = (0, 1, 2, 2)$.
3. Therefore $u = u_2 = (0, 1, 1, 0)$ and $w = 2w_1 + 2w_2 = (2, 0, 2, 0)$. Their sum is $(2, 1, 3, 0)$, as required.